
FastPartz

Solution Comparison

4 Approaches to Securing Your Network

A: OPNsense

B: FlexiWAN

C: Cloud Hub

D: Cloudflare Zero Trust

Date: February 2026

Scope: 4 branches (UAE)

Purpose: Compare cost, complexity, security, and maintenance

Solution A: OPNsense Traditional Firewall

OPNsense + Manual IPsec + Suricata IDS

A dedicated OPNsense firewall appliance (N100 mini-PC) at each branch, with manually configured IPsec tunnels creating a full-mesh VPN between all branches. Each device runs Suricata IDS/IPS for intrusion detection and Unbound DNS for malware/phishing filtering.

Key Characteristics

Aspect	Detail
Edge Device	OPNsense on Intel N100 mini-PC (dedicated per branch)
Inter-Branch	6 manual IPsec tunnels (IKEv2, AES-256-GCM, certificate auth)
Firewall	Full stateful firewall with per-port, per-VLAN rules
IDS/IPS	Suricata with ~35,000 ET Open + Abuse.ch rules, inline blocking
DNS Filtering	Unbound with DNSBL blocklists, DNSSEC, DNS-over-TLS
Remote Access	IKEv2 VPN with certificate + password + TOTP (MFA)
Management	Per-device web GUI (4 separate logins) — no central dashboard
Dynamic DNS	Dynu DDNS on each branch for tunnel peer discovery

Strengths & Weaknesses

Strengths

- Deepest security (IDS/IPS + stateful FW)
- Full open-source, zero recurring cost
- Mature, battle-tested platform
- Massive community + documentation
- Complete feature set (VPN, DNS, DHCP, CA)

Weaknesses

- 4 separate GUIs — no central management
- 12 manual IPsec configurations
- Requires networking expertise
- DynDNS adds fragility
- Cannot be managed remotely without VPN first

Solution A: Architecture Diagram

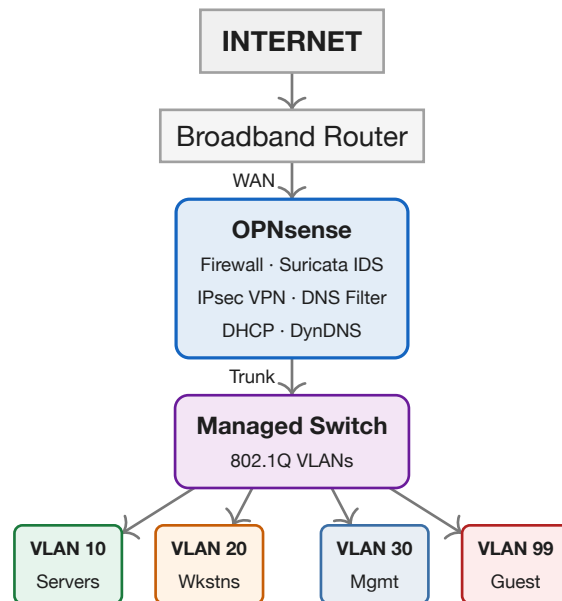


Figure 1: Solution A: Per-branch topology (× 4 branches)

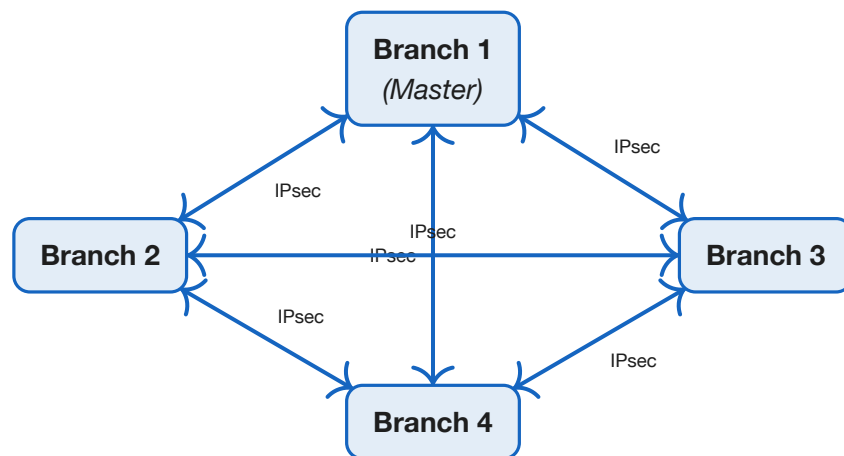


Figure 2: 6 manual IPsec tunnels — full mesh, manually configured on each end

Solution B: FlexiWAN Cloud-Managed SD-WAN

FlexiWAN + Cloudflare Gateway DNS

FlexiWAN SD-WAN appliance at each branch, managed from a single cloud dashboard (flexiManage). Tunnels are created with one click from the dashboard — no manual IPsec configuration. DNS security provided by Cloudflare Gateway (free tier, 50 users).

Key Characteristics

Aspect	Detail
Edge Device	FlexiWAN (Ubuntu 20.04) on Intel N100 mini-PC
Inter-Branch	6 auto-configured IPsec/VXLAN tunnels via flexiManage dashboard
Firewall	Basic iptables + application identification
IDS/IPS	Not available — gap filled by Cloudflare Gateway DNS + endpoint AV
DNS Filtering	Cloudflare Gateway (cloud-based, free for 50 users)
Remote Access	WireGuard VPN on the FlexiWAN device
Management	Single cloud dashboard (flexiManage) for all 4 branches
Dynamic DNS	Not needed — flexiManage handles tunnel peer discovery

Strengths & Weaknesses

Strengths

- Single dashboard for all branches
- One-click tunnel creation (full mesh)
- True SD-WAN (app identification, QoS)
- Push updates to all devices at once
- Remote management from anywhere

Weaknesses

- No IDS/IPS (Suricata equivalent)
- Basic firewall (not stateful NGFW)
- \$160/month cloud management fee
- Requires DPDK-compatible NICs (I226)
- Smaller community than OPNsense

Solution B: Architecture Diagram

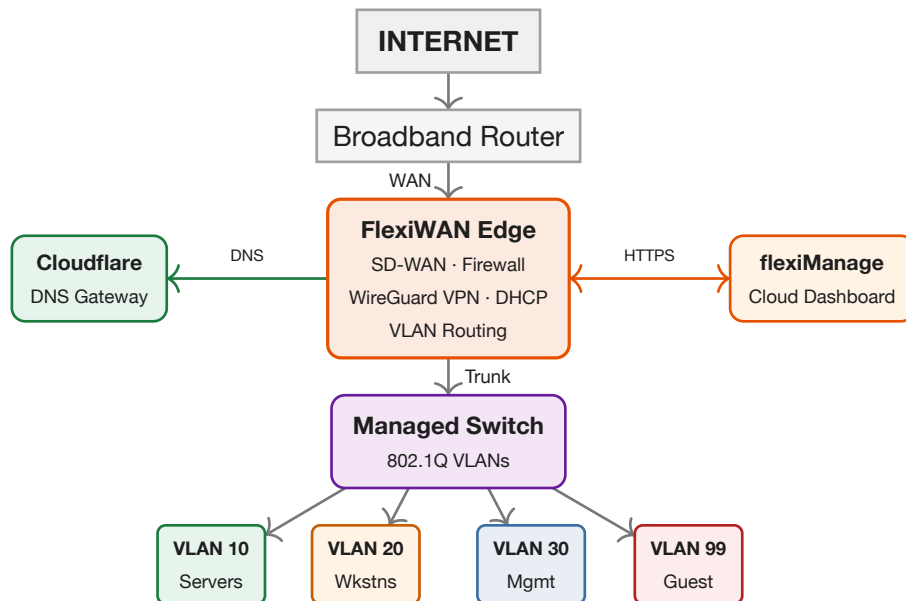


Figure 3: Solution B: Per-branch topology with cloud management + DNS security

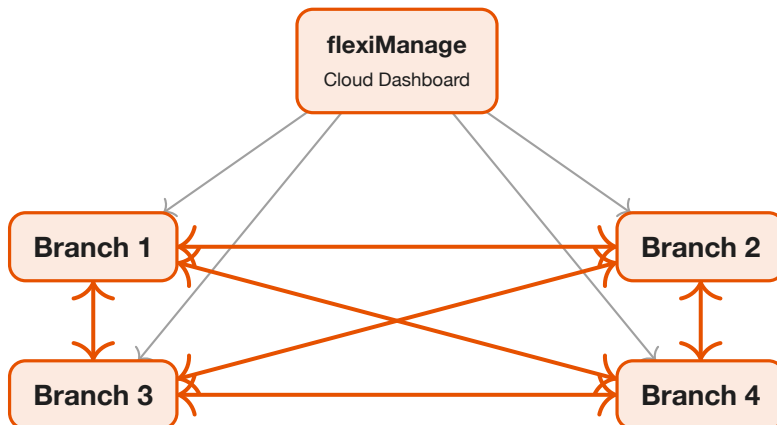


Figure 4: Auto-configured full mesh + cloud management

Solution C: Cloud Hub (OCI UAE)

Central Cloud Server + Outbound Sync Only

A central Windows/Linux server in Oracle Cloud (UAE region) acts as the sync hub. Each branch pushes data outbound to the cloud — no branch talks to any other branch. Remote users RDP to the cloud server. Zero inter-branch tunnels.

Key Characteristics

Aspect	Detail
Cloud Server	OCI VM (Dubai/Abu Dhabi) — Windows Server + SQL or Linux + SQL Express
Inter-Branch	None — all sync goes through the cloud hub
Branch Firewall	OPNsense (simplified — block all inbound, allow outbound to cloud)
DNS Filtering	OPNsense Unbound with DNSBL blocklists
Remote Access	RDP to cloud server (via RD Gateway / HTTPS)
Management	OCI Console + RDP to cloud server — all remote
Sync Model	Branches initiate outbound SQL sync to cloud. Cloud redistributes.

Strengths & Weaknesses

Strengths

- Zero inter-branch tunnels
- Simplest branch firewall config
- Cloud = built-in disaster recovery
- No branch-to-branch lateral movement
- OCI UAE keeps data in-country

Weaknesses

- Monthly cloud cost (\$55–100/mo)
- Accounting software must support hub sync
- Internet dependency for all sync
- Added latency for SQL queries
- Cloud server = new attack surface to protect

Solution C: Architecture Diagram

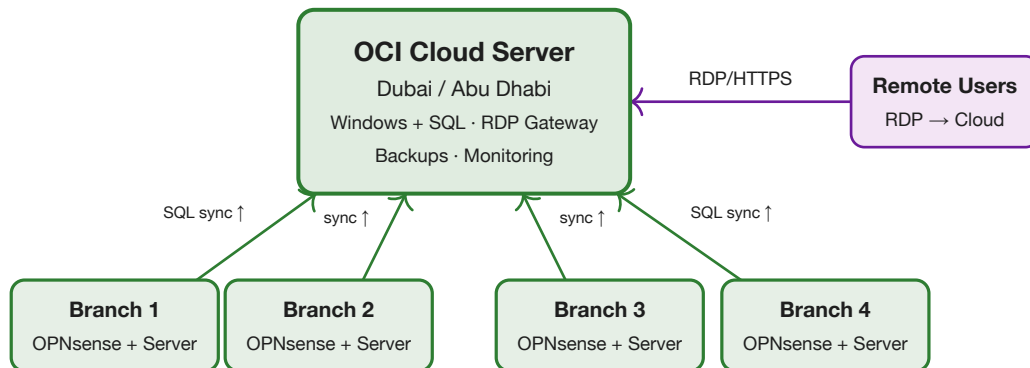


Figure 5: All branches sync outbound to cloud. No branch-to-branch traffic. Remote users RDP to cloud.

Key architecture decision: No branch communicates with any other branch. If Branch 2 is compromised, there is **no network path** to Branch 3. The cloud server is the only shared resource, protected by OCI Security Lists (IP whitelisting) and RD Gateway.

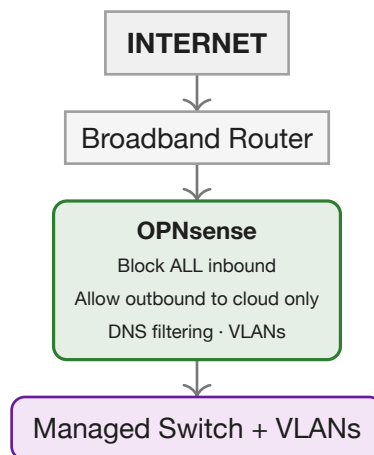


Figure 6: Simplified branch topology — no tunnels, no VPN, just firewall + VLANs

Solution D: Cloudflare Zero Trust Mesh

cloudflared Tunnels + Browser RDP + Free

A lightweight agent (`cloudflared`) installed on each Windows Server creates outbound tunnels to Cloudflare's global network using HTTP/2 or QUIC — indistinguishable from regular HTTPS browsing. Cloudflare routes traffic between branches, provides browser-based RDP for remote users, and enforces zero-trust access policies. No VPN protocols. No cloud server. Free for up to 50 users.

Key Characteristics

Aspect	Detail
Tunnel Agent	cloudflared on each Windows Server (30MB, runs as service)
Protocol	HTTP/2 or QUIC — looks like HTTPS to ISPs. No VPN signatures.
Inter-Branch	Cloudflare routes SQL traffic between branch tunnels
Remote Access	Browser-based RDP — no client software, MFA via M365/Google
DNS Filtering	Cloudflare Gateway (same dashboard, free)
Access Policies	Zero-trust: "Branch A → Branch B on port 1433 only"
Management	Cloudflare Zero Trust dashboard (single pane of glass)
Cost	\$0 — free tier supports up to 50 users

Strengths & Weaknesses

Strengths

- Completely free (Cloudflare Zero Trust)
- Zero hardware beyond basic firewall
- No VPN — uses HTTPS (ISP-invisible)
- Browser-based RDP with MFA
- Granular zero-trust policies
- Single management dashboard

Weaknesses

- Depends on Cloudflare (third-party)
- cloudflared adds 5–20ms latency
- SQL sync through Cloudflare needs testing
- Browser RDP may be slower than native
- Free tier: 24-hour log retention only
- Newer technology — less battle-tested

Solution D: Architecture Diagram

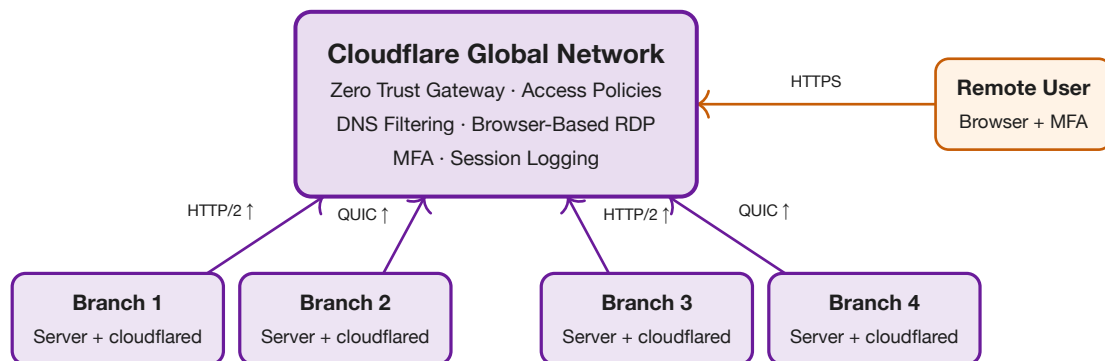


Figure 7: All tunnels outbound (looks like HTTPS). Cloudflare routes traffic + enforces zero-trust policies.

How SQL sync flows: Branch 1 Server → cloudflared (HTTP/2 outbound) → Cloudflare Edge → cloudflared (Branch 2) → Branch 2 SQL Server. To the ISP, this is indistinguishable from regular HTTPS browsing. No VPN signatures.

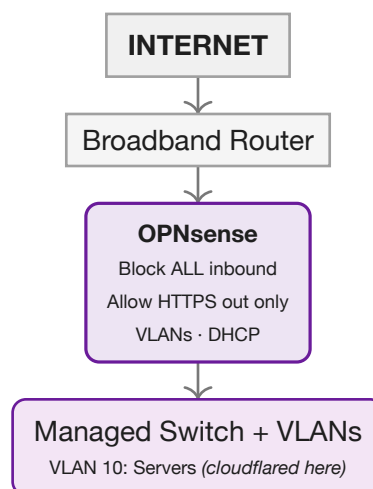


Figure 8: Simplest branch setup — firewall blocks everything, cloudflared handles connectivity

Side-by-Side Comparison

	A: OPNsense	B: FlexiWAN	C: Cloud Hub	D: Cloudflare ZT
Hardware Cost	\$1,520–2,200	\$1,520–2,200	\$1,520–2,200	\$1,520–2,200
Year 1 Recurring	\$160 (ESET only)	\$2,080 (FlexiWAN + ESET)	\$920–1,460 (OCI + ESET + B2)	\$220 (ESET + B2 only)
Year 1 Total	\$1,680–2,360	\$3,600–4,280	\$2,440–3,660	\$1,740–2,420
Year 2+ Annual	\$160	\$2,080	\$920–1,460	\$220
Complexity	HIGH 4 GUIs, 12 IPsec con- figs, DynDNS	MEDIUM 1 dashboard, auto tun- nels	LOW No tunnels, simple FW	LOWEST Software only, 1 dash- board
Security Depth	HIGHEST IDS/IPS + stateful FW + DNS	MEDIUM Basic FW + DNS (cloud)	MEDIUM FW + cloud isolation	HIGH Zero-trust + DNS + MFA
Maintenance	HIGH Per-device, manual tunnels	LOW Cloud-managed, auto- healing	LOW Manage cloud server only	LOWEST Cloudflare manages infra
Remote Admin	Need VPN to reach each FW GUI	flexiManage dash- board (anywhere)	OCI Console + RDP (anywhere)	CF dashboard + browser RDP
Inter-Branch	6 manual IPsec tunnels	6 auto SD-WAN tun- nels	None — cloud hub sync	Cloudflare mesh (HTTPS)
VPN Protocol	IKEv2 IPsec	IPsec/VXLAN + Wire- Guard	None (RDP to cloud)	None (HTTP/2 / QUIC)
IDS/IPS	Suricata (~35K rules)	None	OCI Cloud Guard	CF Gateway (DNS layer)
Lateral Movement Risk	Low (VLANs + tunnels exist)	Low (VLANs + tunnels exist)	None (no branch links)	None (zero-trust poli- cies)
Third-Party Depen- dency	None (self-hosted)	FlexiWAN company	Oracle Cloud	Cloudflare
SW Compatibility Risk	None — same sync model	None — same sync model	Must verify hub sync works	Must test SQL via CF tunnel
Best For	Max security, IT team available	Central mgmt, willing to pay	Simplicity, if hub sync works	Lowest cost + zero VPN needed

Recommendation: For FastPartz (no IT team, remote admin from outside UAE), **Solution D (Cloudflare Zero Trust)** offers the best balance of cost, simplicity, and security — provided SQL sync through Cloudflare tunnels is validated during a pilot phase. **Solution C (Cloud Hub)** is the safest fallback if the accounting software requires a traditional hub-and-spoke sync model.